Data-Driven Optimization of IT Support Team Performance Using Analytics (Supportlytics)

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1. Introduction

In modern organizations, IT support systems generate large volumes of ticket data daily. Efficient handling of these tickets is essential to maintain service quality and operational efficiency.

This project focuses on analyzing IT support ticket data to:

- Identify performance trends
- Optimize resolution time
- Improve resource allocation
- Enhance overall service efficiency

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2. Problem Statement

The objective of this project is to:

- Analyze IT support ticket data
- Identify key performance trends
- Detect issue patterns and clusters
- Optimize resolution time
- Provide actionable insights for improving IT support operations

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3. Dataset Overview

- **Source:** Internal IT Support System

- **Data Type:** Structured CSV
- **Key Features:**
  - Ticket ID
  - Issue Type (Request / Incident / Problem)
  - Priority (Low / Medium / High)
  - Category (Bug, Security, Integration, etc.)
  - Resolution Time
  - Country
  - Cluster ID & Similarity Score

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4. Milestone-wise Implementation

Milestone 1: Data Preparation

Module 1: Project Initialization & Dataset Setup

- **Loaded dataset using pandas**
- **Explored:**
  - Data types
  - Missing values
  - Dataset structure
- **Identified KPIs:**
  - Ticket volume
  - Resolution time
  - Priority distribution

Module 2: Data Cleaning & Feature Engineering

- **Handled missing and inconsistent data**
- **Created new features:**
  - Resolution_Duration = Resolution Date – Created Date
  - Priority Score (for weighted analysis)
- **Encoded categorical variables**
- **Generated cleaned dataset**

Deliverables

- Cleaned dataset
- Feature engineering summary
- Data dictionary

Milestone 2: Visualization & Cluster Analysis

Module 3: Exploratory Data Analysis (EDA)

- **Visualized:**
  - Ticket distribution by type
  - Priority-wise ticket count
  - Category frequency
- **Identified:**
  - Most frequent issue types
  - High-volume categories

Module 4: Similarity & Cluster Insights

- Analyzed clusters using:

    - Cluster_ID
    - Similarity_Score
- **Compared:**
    - Cluster size vs issue type
- **Visualizations used:**
    - Bar charts
    - Boxplots
    - Scatter plots

Deliverables

- 8+ visualizations
- Key insights summary
- Identified problem clusters

Milestone 3: Performance & Geographic Insights

Module 5: Performance Trend Analysis

- **Analyzed:**
    - Average resolution time by priority
    - Ticket handling efficiency
- **Identified:**
    - High-priority delays
    - Slow-performing areas

Module 6: Geographic & Category Analysis

- **Created:**

    - Heatmaps of ticket distribution
    - Geo-maps using latitude & longitude
- **Analyzed:**
    - Region-wise performance
    - Category-wise issue concentration

Deliverables

- Performance metrics summary
- Geo-visualizations
- Comparative analysis charts

Milestone 4: Dashboard & Documentation

Module 7: Dashboard Development

Developed interactive dashboard using:

- Streamlit / Power BI

Key KPIs Displayed

- Average Resolution Time
- Most Frequent Categories
- Cluster Similarity Index
- Top Performing Regions

Module 8: Documentation & Presentation

- **Created:**
    - Final project report
    - Presentation slides

- **Uploaded:**
  - Code
  - Dataset
  - Dashboard
  - Documentation to GitHub

Deliverables

- Interactive dashboard
- Final report (PDF)
- PPT presentation

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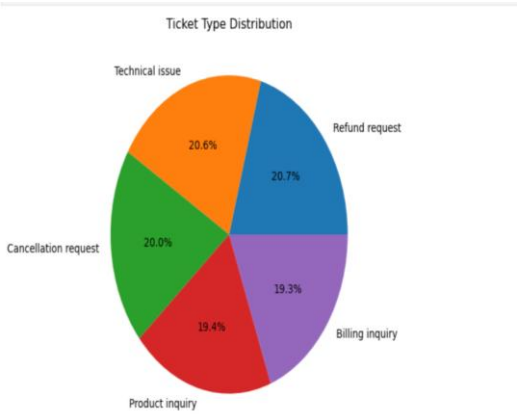
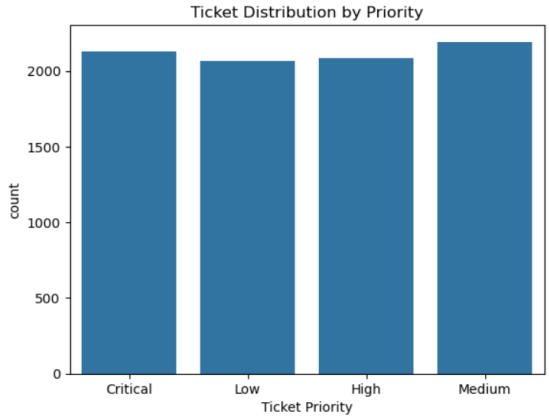
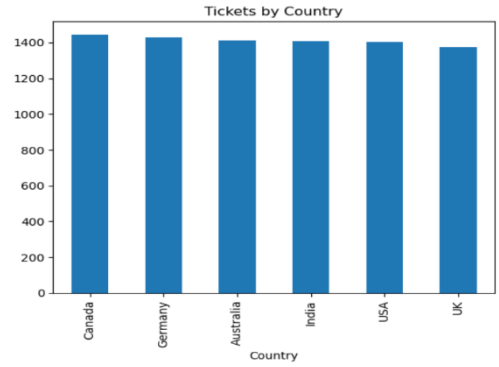
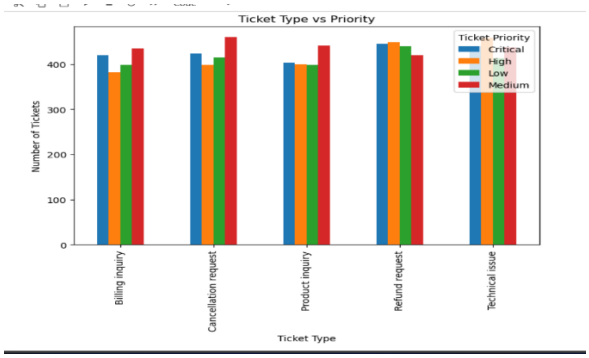
5. Results

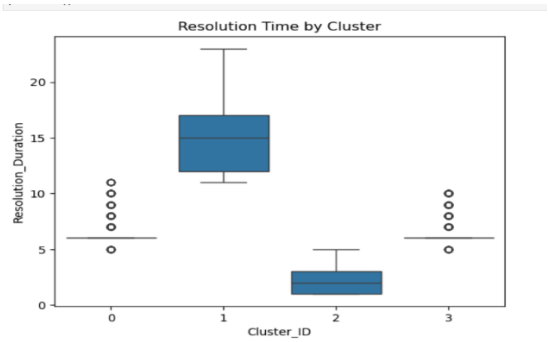
Visualization and Dashboard Development

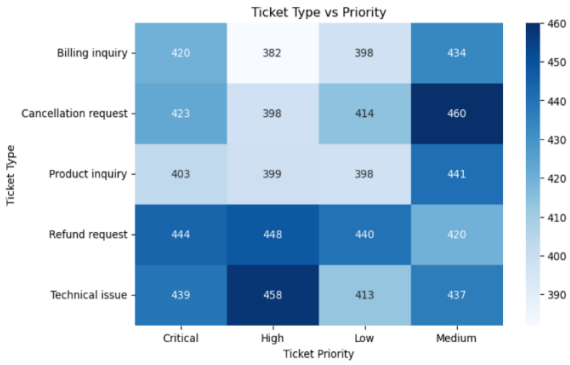




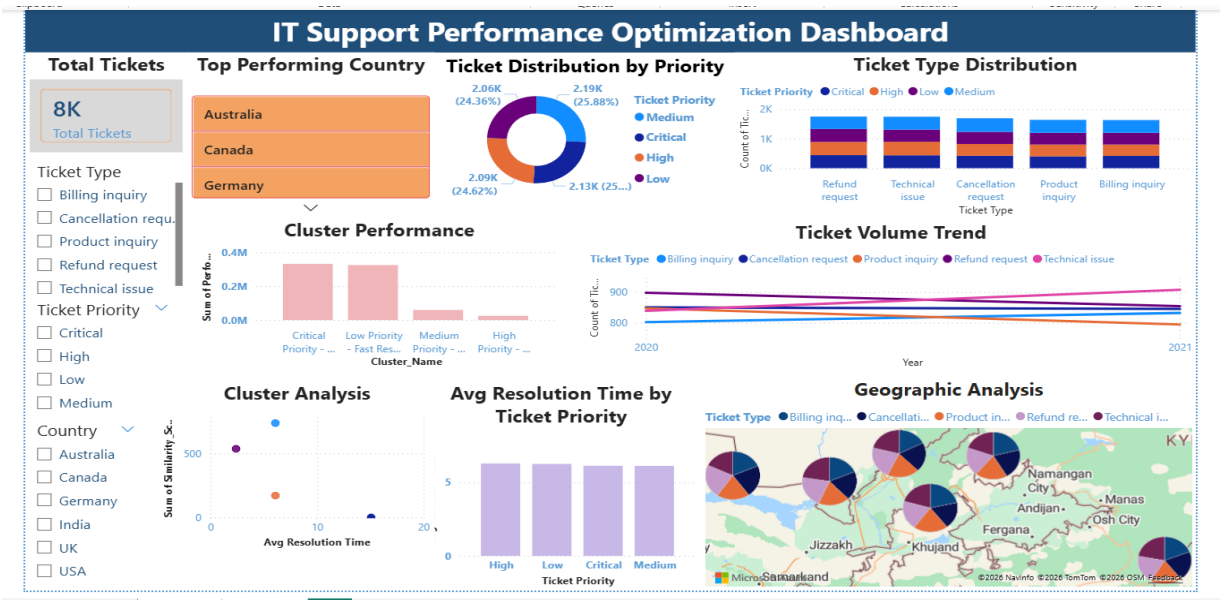








POWER BI DASHBOARD



Key Insights

- High-priority tickets take longer resolution time in some regions
- Certain categories (e.g., bugs, integration issues) dominate ticket volume
- Clustering reveals repeated issue patterns
- Some regions consistently perform better than others
- Resource allocation is uneven across teams

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6. Recommendations

- Allocate more resources to high-priority ticket handling
- Automate repetitive issue categories
- Improve response time for critical clusters
- Balance workload across regions
- Implement predictive analytics for ticket handling

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7. Tech Stack

- Data Processing: pandas, numpy
- Visualization: matplotlib, seaborn, plotly
- Dashboard: Streamlit / Power BI

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8. Conclusion

This project successfully demonstrates how data analytics can improve IT support operations. By analyzing ticket data, identifying patterns, and building dashboards, organizations can:

- Improve efficiency
- Reduce resolution time
- Enhance customer satisfaction

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9. Future Scope

- Machine Learning for ticket prediction
- Automated ticket classification
- Real-time dashboard integration
- AI-based support systems